

Understanding Lasers and Their Applications

Lasers are wonderful devices with enormous application potential. This article gives you an insight into their basic characteristics, classification and applications

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The word 'LASER,' acronym for 'Light Amplification by Stimulated Emission of Radiation,' is very common in the 21st century. Ever since the invention of the first laser (ruby laser) by Theodore Maiman in 1960, it has become one of the necessities today. Moreover, it is no less than a surprise that laser is being extensively used nowadays in both weaponry (missiles, guns, etc) and civilian applications with the same degree of effectiveness. However, despite its vast usage in almost every field, we still have not exploited its potential to the fullest.

In 1917, Albert Einstein established the theoretic foundations for the laser in a paper on quantum theory of radiation, via a re-derivation of Max Planck's law of radiation, conceptually based upon probability coefficients (Einstein coefficients) for absorption, spontaneous emission and stimulated emission of electromagnetic radiation.

Definition and characteristics

Laser is technically defined as an optical device that produces an intense monochromatic beam of coherent light. The light produced by the laser is like any other light, i.e., an electromagnetic wave. But apart from the properties of any beam of light, it also has some other distinct characteristics that differentiate it from ordinary light and make it so valuable and useful (Fig. 2). These characteristics include monochromaticity, coherence, directionality and high intensity.

Monochromaticity. A beam is



Fig. 1: Laser beam of different wavelengths

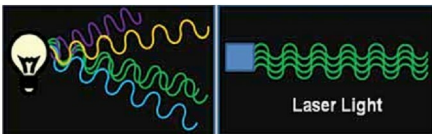


Fig. 2: Radiation from bulb and laser light

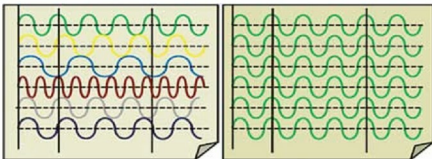


Fig. 3: Incoherent and coherent waves

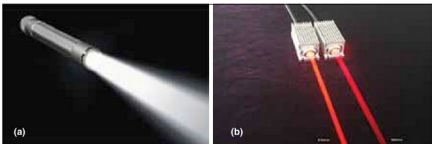


Fig. 4: (a) Beam divergence of a torch, (b) Beam divergence of laser

said to be monochromatic if it consists of a single wavelength. The energy of a photon determines its wavelength through the relationship $E = hc/\lambda$, where 'c' is the speed of light, 'h' is Planck's constant and ' λ ' is the wavelength. Ideally, the laser emits all photons with the same energy, and thus the same wavelength. Hence it is said to be monochromatic. In simple language, light emitted